

ECON 3040 – Intro to Econometrics

Lecture 1 – Course outline, RStudio, “What is Econometrics?”

Course Description

The principal objective of this course is to provide a basic introduction to econometric theory and its application. Much of the emphasis of the course is on the linear multiple regression model, under standard assumptions. The course begins with a review of probability and statistics, and ordinary least squares (OLS).

Required Textbook

[Godwin, R. T., *Introduction to Econometrics*](#)

Recommended Textbook

Introduction to Econometrics, 3rd Edition Update, by Stock and Watson.

Course Website

Course resources (including lecture notes, past exams, assignments, and computer labs) are available on rtgodwin.com/3040

Evaluation

Assignments:	10%
Midterm 1 (Feb. 3):	22.5%
Midterm 2 (Mar. 10):	22.5%
Final Exam:	45%

Assignments

You will use RStudio and work with data in order to complete your assignments.

Midterm and final examination

These will be closed book/closed notes. The final examination will cover all of the material presented in the course.

Grading scale

A+	93 – 100
A	87 – 93
B+	80 – 87
B	72 – 80
C+	64 – 72
C	57 – 64
D	50 – 57
F	0 – 50

- A missed assessment will result in make-up work, or reweighting of your grade.
- Nov. 24 is the last day for Voluntary Withdrawal from courses.

Academic Integrity

- All assignments and exams must be completed independently.
- Do not engage in “contract” cheating.
- Do not provide your UM Learn login information to anyone else. This is “personation”, a serious form of academic misconduct.

Ignorance is not a defense. Familiarize yourself with section 2.5 of [Academic Misconduct Procedures](#).

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Tentative Course Topics

- Review of Probability
- Review of Statistics
- Linear Regression with One Regressor
- Hypothesis Tests
- Linear Regression with Multiple Regressors
- Hypothesis Tests in Multiple Regression
- Nonlinear Regression Functions
- Instrumental Variables
- Heteroskedasticity

Student Accessibility Services

Students with disabilities should contact Student Accessibility Services to facilitate the implementation of accommodations, and meet with me to discuss the accommodations recommended by Student Accessibility Services.

Academic Supports

Sample Lecture

What is Econometrics?

- Econometrics is a subset of statistics
- Science of testing economic theories
- Used to estimate causal effects
- Used to forecast or predict (not covered in this course)
- Often characterized by “observational data”

Causal Effects

Economic models often suggest that one variable causes another. This often has *policy implications*. The economic models, however, do not provide *quantitative magnitudes* of the causal effects.

For example:

- How would a change in the *price* of alcohol or cigarettes effect the *quantity* consumed?
- If *income* increases, how much of the increase will be *consumed*?
- If an additional *fireplace* is added to a house, how much will the *price* of the house increase?
- How does another year of *education* change *earnings*?

Using data to estimate causal effects

An experiment would be best.

- How would you determine the effect of fertilizer on crop yield?
- How would you use an experiment to determine the above four causal effects (on the previous slide)?
- What is the advantage of experiments?

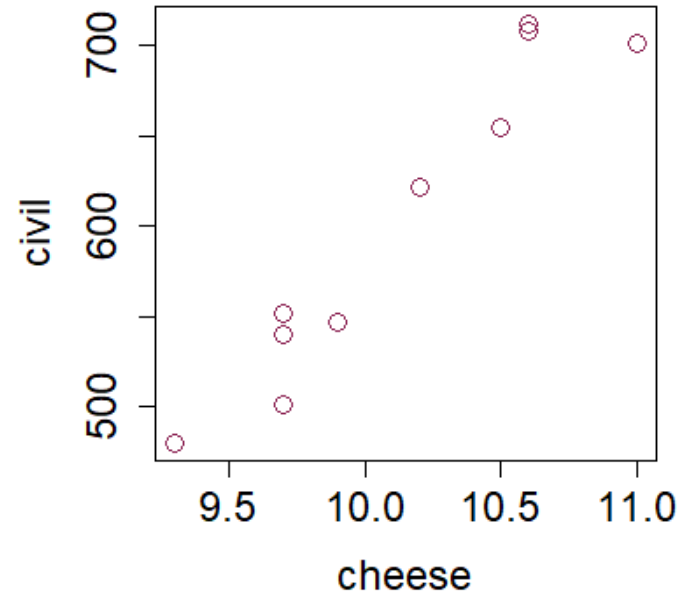
Economic experiments are usually unethical and/or too expensive.

We usually don't have *experimental* data in econometrics – we have *observational* data.

There are issues when dealing with observational data:

- Omitted variables
- Simultaneous causality
- Correlation vs. causation

Civil engineering PhDs awarded, and per-capita consumption of cheese, from 2000-2009 in the U.S. (Spurious correlations, Tyler Vigen)



What is wrong with the above picture?

Objectives of this course

- Learn a method for estimating causal effects (least squares, “LS”)
- Understand some theoretical properties of LS
- Learn about hypothesis testing
- Practice LS using data sets

R and RStudio

The theory and concepts presented in this course will be illustrated by analysing several data sets. Data analysis will be accomplished through the R Statistical Environment and RStudio. Both are free, and R is fast becoming the best and most widely used statistical software.

First, install R

- Go to <https://muug.ca/mirror/cran/>
- Choose Windows or Mac

← → ↻ muug.ca/mirror/cran/ ☆ ⚙ R ⋮



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The Comprehensive R Archive Network

Download and Install R

Precompiled binary distributions of the base system and contributed packages, **Windows and Mac** users most likely want one of these versions of R:

- [Download R for Linux](#)
- [Download R for \(Mac\) OS X](#)
- [Download R for Windows](#)

R is part of many Linux distributions, you should check with your Linux package management system in addition to the link above.

Source Code for all Platforms

Windows and Mac users most likely want to download the precompiled binaries listed in the upper box, not the source code. The sources have to be compiled before you can use them. If you do not know what this means, you probably do not want to do it!

- The latest release (2020-06-22, Taking Off Again) [R-4.0.2.tar.gz](#), read [what's new](#) in the latest version.
- Sources of [R alpha and beta releases](#) (daily snapshots, created only in

- Click “install R for the first time”

← → ↻ muug.ca/mirror/cran/ ☆ ⚙ R ⋮



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R for Windows

Subdirectories:

base	Binaries for base distribution. This is what you want to install R for the first time .
contrib	Binaries of contributed CRAN packages (for R >= 2.13.x; managed by Uwe Ligges). There is also information on third party software available for CRAN Windows services and corresponding environment and make variables.
old contrib	Binaries of contributed CRAN packages for outdated versions of R (for R < 2.13.x; managed by Uwe Ligges).
Rtools	Tools to build R and R packages. This is what you want to build your own packages on Windows, or to build R itself.

Please do not submit binaries to CRAN. Package developers might want to contact Uwe Ligges directly in case of questions / suggestions related to Windows binaries.

You may also want to read the [R FAQ](#) and [R for Windows FAQ](#).

Note: CRAN does some checks on these binaries for viruses, but cannot give guarantees. Use the normal precautions with downloaded executables.

- Click “Download R 4.4.1 for Windows” (or Mac)
- Run the “.exe” file
- Click “Next” a bunch of times
- Don’t download RTools!

Second, install RStudio

- Go to <https://rstudio.com/products/rstudio/download/>
- Scroll down until you see the download button “Download RStudio Desktop for Windows (Mac)”. Click it.

Step 2: Install RStudio Desktop

DOWNLOAD RSTUDIO DESKTOP FOR WINDOWS

Size: 202.76MB | [SHA-256: FD8EA4B4](#) | Version: 2022.12.0+353 |
Released: 2022-12-15

- Run the “.exe”
- Keep clicking “next” / “install”
- Find RStudio on your computer and open it. It should look something like this:

